UNIVERSITY AT ALBANY
STATE UNIVERSITY OF NEW YORK

Massry School of Business
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from opportunity to impact

BFOR516 – Advanced Data Analytics (3 Cr.)
Spring 2026, Class #7802
Online, Tuesday 6:00pm – 8:50pm

Instructor: Dr. Srishti Gupta, Visiting Assistant Professor, Information Security and Digital Forensics
Email: sgupta4@albany.edu
Office Location: BB 318
Office Hours: Wednesday 10:00 am – 11:00 am, Thursday 11:00 am - 12:00 pm

Description:

This course provides an introduction to data analytics techniques and their applications, with a focus on enabling students to develop and apply machine learning models to solve complex problems in cybersecurity and digital forensics. Students will explore a range of methods, including dimensionality reduction, supervised and unsupervised learning, neural networks, and generative AI. Through lectures, hands-on exercises, and case studies, the course emphasizes practical applications, feature engineering, and critical evaluation of data analytics research.

Students will work collaboratively in teams to analyze research papers as structured, problem-based scenarios. These activities require students to critically evaluate analytical methods, interpret findings, and reflect on broader implications for cybersecurity and digital forensics, mirroring real-world problem-solving processes. By the end of the course, students will have a comprehensive understanding of data analytics fundamentals, ethical considerations, and the skills needed to effectively integrate data analytics solutions into cybersecurity contexts.

Overview and Objectives:

This course is designed to build foundational and applied competencies in machine learning and artificial intelligence for cybersecurity and digital forensics. Upon successful completion of the course, students will be able to:

1. Understand core concepts of machine learning and artificial intelligence
2. Apply feature engineering techniques to security and forensic datasets
3. Explore and evaluate neural network architectures
4. Apply problem-based reasoning to real-world cybersecurity challenges

5. Integrate generative AI tools responsibly to plan, debug, and reflect on analytical solutions
6. Understand the ethical and practical implications of data analytics
7. Collaborate effectively and communicate findings within interdisciplinary teams

Assurance of Learning (AOL) Objectives:

This course contributes to the School's Assurance of Learning (AOL) process in support of AACSB accreditation. BFOR 516 is a Course Assessed for the following learning objective:

- Students will possess competencies in data analytics that they can leverage in specialization courses.

Student achievement of this AOL objective will be evaluated using curriculum-embedded assessments, such as applied machine learning assignments, team-based research analysis, and problem-based projects focused on cybersecurity and digital forensics.

Prerequisites: BFOR 515. Students who have received credit for BFOR416 may not receive credit for this course.

Grading:

The grade system for the course is from A-E, and will follow the following grading scale:

Grade Scale	Grade Conversion	Grade Scale	Grade Conversion
93-100	A	73-76	C
90-92	A-	70-72	C-
87-89	B+	67-69	D+
83-86	B	63-66	D
80-82	B-	60-62	D-
77-79	C+	Grade < 60	E

Grades will be weighted based on the following:

Activity	Percentage of Grade
In-Class Lab Work	20%
Individual Critical Analysis Paper	30%
Group Project	25%

Individual Learning Portfolio	15%
Attendance	10%
Total	**100%**

All submissions will be graded fairly, and grades posted within a reasonable timeframe. I will be available to discuss questions, provided you set up an appointment within a week of receiving a grade. Please let me know if you feel an error in calculation has been made - mistakes do happen!

Submitting late assignments, projects or papers will result in points being deducted from the final possible grade, unless there is a verifiable legitimate excuse or approved extension. Each day of late submission will result in a 10% grade deduction. Missing an assignment completely will obviously result in a grade of zero.

Textbooks for Reference:

1. **James, Witten, Hastie, & Tibshirani - *An Introduction to Statistical Learning***
   James, G., Witten, D., Hastie, T., & Tibshirani, R. (2023). *An introduction to statistical learning: With applications in Python*. Springer. https://www.statlearning.com
2. **Géron - Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow**
   Géron, A. (2019). Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems (2nd ed.). O'Reilly Media.
3. **Goodfellow, Bengio, & Courville - Deep Learning**
   Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press. https://www.deeplearningbook.org

Course Delivery:

This course will be delivered synchronously online. The format is designed to offer an interactive learning experience, incorporating lectures, discussions, in-class activities, and collaboration among peers and the instructor. The course will utilize Brightspace as the central platform for distributing course materials, facilitating online discussions, and submitting assignments. The overall evaluation will be based on four key components:

1. **In-Class Lab (20%)**: Students will engage in small lab activities during each class session, directly based on that day's lecture. These activities are designed to provide hands-on experience and reinforce the concepts discussed in class.
2. **Individual Critical Analysis Paper (30%):** This is an individual assignment which will be due in the middle of the semester. Students will analyze and write a critical review of a research paper that aligns with one of the course's key topics. Students will select one research paper from an instructor-provided list. Each student will submit an individual critical analysis of the selected paper based on instructor guidelines. Although multiple students may analyze the same paper, all submissions must reflect independent reasoning and original analysis. Detailed instructions and deadlines will be provided on Brightspace.
3. **Group Project (25%):** Student groups will collaboratively analyze 2-3 machine learning models applied to a common dataset or problem scenario. The project emphasizes performance comparison, interpretability, and ethical considerations. Groups will submit a project report and deliver a final

presentation summarizing their findings. Each group must justify their model selection, evaluation approach, and interpretation. Detailed instructions and deadlines will be provided on Brightspace.

4. **Individual Learning Portfolio (15%)**: Each student will submit a short reflective portfolio documenting their learning throughout the course. The portfolio should include:
    a. Key learning moments and challenges encountered
    b. How generative AI tools were used, evaluated, and integrated into your work
    c. Reflections on what you would do differently in future analyses or projects
5. **Attendance (10%):**
    a. Attendance will be taken each class.
    b. Students must attend the entire class to receive credit. Partial attendance will not earn points.
    c. This ensures engagement and participation.

Course Schedule:

Week	Date	Topic	Description
Week 1	January 27	Introduction to class and recap of basic concepts	Introduction to class policies and schedule, recap of basic concepts like regression analysis
Week 2	February 3	Dimensionality Reduction	Principle Component Analysis
Week 3	February 10	Supervised Learning	Logistic Regression, Decision Trees, Naïve Bayes
Week 4	February 17	Unsupervised Learning	Hierarchical Clustering, K- means, DBSCAN
Week 5	February 24	Representing Data and Engineering Features	Feature Engineering, Feature Selection
Week 6	March 3	Introduction to Neural Networks	Artificial Neural Networks key components, Backpropagation,
Week 7	March 10	Types of NNs	Convolutional Neural Networks
Week 8	March 17	NO CLASS	SPRING BREAK
Week 9	March 24	Types of NNs	Recurrent Neural Networks
Week 10	March 31	Modern RNNs	LSTM, GRU, and Attention Mechanism
Week 11	April 7	Transformers and LLMs	Encoder-Decoder, Multi-Head Attention, Self-Attention, Transformer Architecture
Week 12	April 14	Prompt Engineering	Foundational Concepts, Prompting Techniques
Week 13	April 21	AI Ethics	Ethical Implications of AI, XAI
Week 14	April 28	Group Presentation	Student group presentations
Week 15	May 5	Group Presentation	Student group presentations and course wrap-up

Attendance and Absence Policies:

Attendance is mandatory, and students are expected to participate in class fully, just as they would in any traditional in-person course. Students must attend the entire class to receive attendance credit.

Students are expected to inform the instructor at least 24 hours prior to class if they anticipate missing a class session. If an emergency arises within 24 hours of the class meeting and prevents attendance, students must notify the instructor as soon as possible and submit appropriate documentation to explain the absence. Failure to provide timely notification and proper documentation may result in the absence being unexcused. For approved excused absences, students will not lose attendance credit for that day; however, they are responsible for making up any missed work, in-class activities, or quizzes within a timeframe specified by the instructor.

Missing class without a valid excuse counts as 0 points for attendance for that session. Repeated unexcused absences (e.g., more than 4) may impact your ability to pass the course.

Policy on Incompletes

Request for an Incomplete will be critically scrutinized for eligibility. Military service, serious ill health, and other such demanding circumstances constitute typical grounds. A request must be made by April 10, unless the event leading to the request occurs after that. Documentation will be required.

Academic Integrity:

Every student has the responsibility to become familiar with the standards of academic integrity at the University. Faculty members must specify in their syllabi information about academic integrity, and may refer students to this policy for more information. Nonetheless, student claims of ignorance, unintentional error, or personal or academic pressures cannot be excuses for violation of academic integrity. Students are responsible for familiarizing themselves with the standards and behaving accordingly, and UAlbany faculty are responsible for teaching, modeling and upholding them. Anything less undermines the worth and value of our intellectual work, and the reputation and credibility of the University at Albany degree. (University's Standards of Academic Integrity Policy, Fall 2013)

Diversity, Equity, and Inclusion (DEI) Statement:

The University at Albany is committed to providing a diverse and inclusive learning, living, and working environment that is free from inequality and discrimination, and to fostering a nurturing and vibrant community founded upon the fundamental dignity and respect of all of its members. The Massry School of Business, in turn, promotes this commitment through coursework, programming, training, mentoring and supporting the needs of students from all backgrounds. Students are encouraged to utilize the University's Office of Diversity & Inclusion and the School's Committee on Inclusion, Equity, and Engagement (CIEE) to learn more about available resources. Massry School of Business students are invited and encouraged to participate in the "THIS IS bUSiness" initiative which spotlights and celebrates our diverse student body.

John S. Levato Career Services Resources:

Whether you are planning for employment or graduate school, take advantage of these School of Business state-of-the-art professional development resources (access them using your UAlbany email address: (1) VMock online resume critique system, and (2) Big Interview-interview practice system. More details are

here.
The School of Business offers career service resources to business and accounting majors, if you are looking for assistance with your job or internship search, resume and/or cover letter assistance, interview preparation or any career related assistance, please email Director, Deirdre Sweeney at dsweeney@albany.edu to set up an appointment or Chris Wessell at chriswessell.com.

Basic Needs Assistance:

It is difficult to succeed academically if you don't have enough to eat, a safe place to live and sleep, or are struggling with an unforeseen emergency. Knowing the resources available on your campus to help you succeed is key! If you need help meeting these or other basic needs, please seek assistance from Supplemental Support Services in the Dean of Students Office. View the basic needs assistance offerings at https://www.albany.edu/dean-students/supplemental-support-services. While you're there, see the variety of helpful services available to you at the Dean of Students at https://www.albany.edu/dean-students. Contact information can be found on these websites.

DAISS Accommodations Information:

Reasonable accommodations will be provided for students with documented physical, sensory, systemic, medical, cognitive, learning and/or mental health (psychiatric) disabilities.

If you believe you have a disability and require accommodation in this class, please register with Disability Access and Inclusion Student Services (DAISS). You can contact DAISS at daiss@albany.edu, 518-442-5501 or https://www.albany.edu/dean-students/disability. Once you have registered with DAISS, they will provide you with an accommodation letter that you can send to your instructors to receive your approved accommodation.